SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1004	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:		**Reg. No.:**	
Date:		**Duration:**	60 minutes

Code of Conduct

*I **M.DEVI SRILATHA (192525236)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ***Srilatha***

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>*.
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

1. Project Vision and Stakeholder Identification

1.1 Project Vision

The vision of the AI-Based Legal Document Management System is to develop a secure and intelligent platform that helps law firms and legal departments store, organize, search, and manage legal documents efficiently. The system uses AI-based document classification, intelligent search, version control, and deadline reminders to reduce manual work and improve productivity.

1.2 Project Objectives

1. To store legal documents securely.
2. To classify documents automatically using AI.
3. To search and retrieve documents quickly.
4. To maintain different versions of documents.
5. To manage legal cases and related documents.
6. To provide reminders for important deadlines.
7. To protect confidential documents through access control.

1.3 Stakeholder Identification

Stakeholder	Role
Lawyer	Uploads, searches, and manages legal documents.
Legal Assistant	Organizes documents and updates case information.
Administrator	Manages users, permissions, and system settings.
Client	Views documents shared with them by authorized users.
Development Team	Designs, develops, tests, and maintains the system.
AI/OCR Service Provider	Supports text extraction and document classification.
Law Firm Management	Monitors efficiency, security, and system usage.

2. Create Epics and User Stories

An epic is a large functionality that can be divided into smaller user stories.

Epic 1: User Authentication and Access Control

User Story 1: Login

As a lawyer, I want to log in securely so that I can access authorized legal documents.

User Story 2: User Permissions

As an administrator, I want to assign user permissions so that confidential documents are accessible only to authorized users.

Epic 2: Document Management

User Story 3: Upload Documents

As a legal assistant, I want to upload legal documents so that they can be stored digitally.

User Story 4: Search Documents

As a lawyer, I want to search documents by name, category, or keyword so that I can retrieve them quickly.

Epic 3: AI-Based Document Processing

User Story 5: Automatic Classification

As a legal assistant, I want the system to classify uploaded documents automatically so that manual categorization is reduced.

User Story 6: OCR Text Extraction

As a lawyer, I want the system to extract text from scanned documents so that I can search their contents.

Epic 4: Version and Case Management

User Story 7: Version Control

As a lawyer, I want to maintain different document versions so that I can view previous changes.

User Story 8: Case Management

As a legal assistant, I want to group documents by case so that all related files are organized together.

Epic 5: Deadline and Notification Management

User Story 9: Deadline Reminders

As a lawyer, I want to receive reminders about important legal deadlines so that I do not miss them.

3. Define Acceptance Criteria

Acceptance criteria describe the conditions that must be satisfied for a user story to be considered complete.

User Story 1: Login

- Given a registered user has valid credentials,
- When the user enters the correct username and password,
- Then the system must allow access to the appropriate dashboard.
- And invalid credentials must produce an error message.

User Story 2: User Permissions

- Given the administrator is logged in,
- When the administrator assigns a role to a user,
- Then the system must save the role and enforce its permissions.
- And unauthorized users must not access restricted documents.

User Story 3: Upload Documents

- Given an authorized user is logged in,
- When the user uploads a supported legal document,
- Then the system must store the file and display a success message.
- And unsupported files must produce an appropriate error message

User Story 4: Search Documents

- Given legal documents are stored in the system,
- When a user searches by document name, category, or keyword,
- Then the system must display matching documents that the user is authorized to access.
- And it must display a suitable message when no results are found.

User Story 5: Automatic Classification

- Given a document has been uploaded,
- When the AI classification process runs successfully,
- Then the system must assign an appropriate document category.
- And the user must be able to review or correct the suggested category.

User Story 6: OCR Text Extraction

- Given a supported scanned document has been uploaded,
- When OCR processing is completed,
- Then the system must extract readable text where possible.
- And the extracted text must be searchable.

User Story 7: Version Control

- Given a document already exists,
- When an authorized user uploads a revised version,
- Then the system must preserve the previous version and save the new version.
- And authorized users must be able to view version history.

User Story 8: Case Management

- Given a legal case exists in the system,
- When an authorized user associates documents with that case,
- Then the documents must appear under the selected case.
- And users must be able to view the case's associated documents according to their permissions.

User Story 9: Deadline Reminders

- Given a legal deadline has been recorded,
- When the configured reminder time is reached,
- Then the system must generate a reminder through the configured notification method.
- And the reminder status must be recorded.

4. Prioritize the Product Backlog

The product backlog contains all planned features, arranged according to importance, business value, and dependencies.

Priority levels:

- High: Essential features required for the initial system.
- Medium: Important features that can follow the basic implementation.
- Low: Additional improvements that can be implemented later.

Backlog ID	User Story / Feature	Priority	Business Value
US1	Secure Login	High	Protects the system from unauthorized access.
US2	User Permissions	High	Protects confidential legal documents.
US3	Upload Documents	High	Enables digital document storage.
US4	Search Documents	High	Reduces document retrieval time.
US5	Automatic Classification	Medium	Reduces manual categorization.
US6	OCR Text Extraction	Medium	Makes scanned documents searchable.
US7	Version Control	Medium	Preserves document history.
US8	Case Management	High	Organizes related legal documents.
US9	Deadline Reminders	Medium	Helps prevent missed deadlines.

Prioritization justification:

Login and access control are prioritized because document security is essential. Uploading and searching are next because they provide the basic document management functionality. AI processing, version

control, and reminders can be developed after the core system is established. Case management is also a high-priority feature because it organizes documents around legal cases.

5. Plan the Sprint Backlog

Sprint Duration: 1 Week

Sprint Number: Sprint 1

Sprint Objective: Develop the basic system foundation with secure login, document upload, and document search.

5.1 Selected User Stories

User Story	Priority	Story Points
US1 – Secure Login	High	3
US3 – Upload Documents	High	5
US4 – Search Documents	High	5
Total		13

5.2 Development Tasks

Task ID	Task Description	Related User Story
T1	Design the login page.	US1
T2	Implement credential validation and authentication.	US1
T3	Create the document upload interface.	US3
T4	Implement file validation and document storage.	US3
T5	Design the document search interface.	US4
T6	Implement keyword-based document search.	US4
T7	Test login, upload, and search functionality.	US1, US3, US4
T8	Fix identified errors and document the results.	US1, US3, US4

Sprint planning note

The team should confirm that 13 story points fit its actual capacity before committing to the sprint. The tasks above are planned for a one-week sprint, assuming a small team and a basic working prototype.

6. Estimate Effort Using Story Points

Story points estimate the relative effort, complexity, and uncertainty of a user story.

The Fibonacci sequence is used for estimation:

1, 2, 3, 5, 8, 13

User Story	Estimated Points	Justification
US1 – Secure Login	3	Requires a login interface, credential validation, and basic authentication testing.
US3 – Upload Documents	5	Requires a user interface, file validation, storage, and error handling.
US4 – Search Documents	5	Requires search input, query processing, result display, and testing.
US5 – Automatic Classification	8	Requires AI integration, category assignment, and accuracy testing.
US6 – OCR Text Extraction	8	Requires OCR integration, text extraction, and handling scanned documents.
US7 – Version Control	5	Requires version storage, version history, and retrieval functionality.
US8 – Case Management	5	Requires case creation, document association, and case-wise retrieval.
US9 – Deadline Reminders	5	Requires deadline storage, reminder scheduling, and notification handling.
US2 – User Permissions	5	Requires role management, access checks, and security testing.

Estimation technique

Planning Poker can be used by team members to discuss the complexity of each story and agree on a final estimate.

Justification

Simple features receive fewer points, while features requiring external services, additional logic, or more testing receive more points. These estimates are initial planning values and may change after technical investigation.

7. Sprint Goal and Expected Deliverables

7.1 Sprint Goal

To develop a basic working version of the AI-Based Legal Document Management System that allows authorized users to log in, upload legal documents, and search stored documents.

7.2 Expected Deliverables

At the end of Sprint 1, the team expects to deliver:

1. A working login page with basic authentication.
2. A document upload interface.
3. Basic document validation and storage.
4. A document search feature.
5. Initial test cases and test results.
6. A basic user interface for demonstrating the features.

7.3 Definition of Done

A user story is considered complete when:

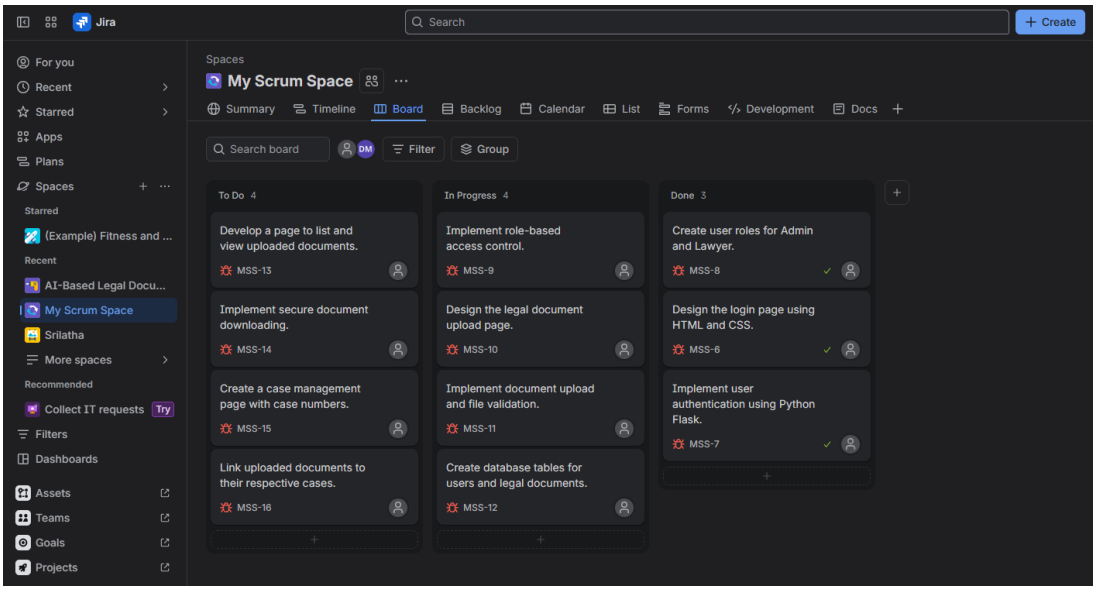
- Its acceptance criteria are satisfied.
- The feature has been tested.
- Identified critical errors have been fixed.
- Unauthorized access is checked where relevant.

- The code and implementation notes are saved.
- The feature works in the integrated prototype.

7.4 Future Sprint Plan

Sprint	Planned Features
Sprint 1	Login, document upload, and document search.
Sprint 2	AI-based document classification and OCR text extraction.
Sprint 3	Version control and case management.
Sprint 4	Deadline reminders, additional security testing, and system improvements.

Screenshots / Output Evidence



Conclusion

The Agile Sprint Planning and User Story Workshop organizes the AI-Based Legal Document Management System into epics, user stories, acceptance criteria, and prioritized backlog items. Story points help estimate the relative effort required for each feature. Sprint 1 focuses on login, document upload, and search to establish a working foundation before adding AI-based processing and advanced management features.